Utility applications and antivirus software

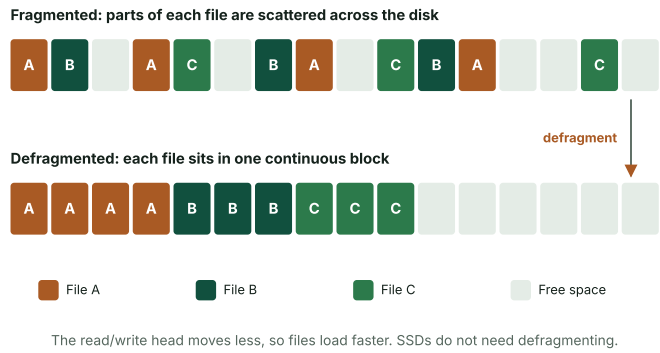
L07 FACT FILE

Specification: students should be able to

- Describe the following tasks carried out by the utility applications: disk defragmenting, task scheduling, backup and restoring data
- Explain why file fragmentation happens and how defragmentation improves performance
- Describe full and incremental backups and a suitable backup procedure
- Describe the role of antivirus software and the importance of regular updates

CCEA fact file: Software

Key words: utility software, data block, file fragmentation, disk defragmenter, read/write head, task scheduling, round robin, backup, restore, full backup, incremental backup, off-site storage, virus, antivirus software, virus signature, quarantine, update



Disk blocks before and after defragmentation

Utility software and defragmentation

Utility software is a group of small programs that each carry out one task to help the operating system keep the computer running well. They include the disk defragmenter, the task scheduler, backup and restore tools and antivirus software. To understand defragmentation, think about how files are saved. The operating system splits a file into blocks and puts each block in any free space on the disk. As files are saved and deleted over months, the free space ends up as lots of small gaps spread around the disk. A large new file may then be split into pieces stored in different places. This is called file fragmentation. On a hard disk drive, the read/write head has to keep jumping around to collect every piece, which slows the computer down. A defragmenter moves the pieces so each file is stored in one continuous area and the free space is grouped together. Solid-state drives have no moving parts, so they gain nothing from this.

Task scheduling

Task scheduling means dividing the processor's time between several tasks so the system is used as efficiently as possible. In a multi-user system, the scheduler uses the round robin method. The processor works on a task from the first computer for a short time slice, then moves to the second, then the third, and after the last one it goes back to the start. Because each slice is tiny and the processor is very fast, everybody feels that they have the computer to themselves. The word task scheduler is also used for the utility that runs jobs automatically at a planned time or when something happens. For example, a school's network manager might schedule a full virus scan every Friday evening, a backup at two in the morning and updates to install when computers shut down. The jobs run when nobody is using the system, so work is not interrupted, and they are never forgotten.

Backup and restore

A backup is a copy of data stored separately from the original so that it can be used if the original is lost or damaged. Data can be lost when a hard disk fails, when a virus or ransomware attacks, when someone deletes a file by mistake, or when a fire or flood destroys a building. For a hospital, a bank or a school, losing records could be disastrous. Backups should be kept off-site or in the cloud, somewhere secure, so a single disaster cannot destroy both copies. A full backup copies all the data, which takes a long time and a lot of space. An incremental backup only copies data that has changed since the last backup, so it is much quicker. Many organisations run a full backup once a week and incremental backups on the other nights, scheduled overnight when the network is quiet. Restoring means copying the backed-up data back onto the system to return it to its earlier state.

Antivirus software

A virus is a harmful program that attaches itself to a file. When the file is opened, the virus runs and spreads to other files and to other computers on a network or the internet. Viruses can damage or delete data, stop a computer from starting up and slow it down. Antivirus software protects against them. It scans the files on the computer's internal drive and on any connected storage, as well as data arriving through emails and downloads. It compares what it finds against a database of known virus signatures, which are patterns of code belonging to each virus. Many programs also look for suspicious behaviour that might come from a brand-new virus. When it finds a threat, the software removes it, repairs the file or moves it into quarantine so it cannot run. Because new viruses appear all the time, the signature database must be updated regularly, otherwise the newest threats will slip through undetected.

Key terms from the mark schemes

Utility software
A program that carries out one specific task to help the operating system maintain, protect or speed up the computer, e.g. defragmenter, task scheduler, backup, antivirus.
Credited: U1 2023 Q5

Disk defragmenter
A utility that rearranges the fragments of files on a disk so each file is stored in continuous blocks and free space is grouped together, speeding up file access.
Credited: U1 2019 Q5, 2022 Q5

Round robin
A scheduling method in which each task or user is given a time slice of the processor in turn, returning to the first after the last.

Restore
Copying data back from a backup to return data or a system to an earlier state.
Credited: U1 2022 Q5

Incremental backup
A backup that copies only the data that has changed since the last backup; quicker and smaller than a full backup.

Antivirus software
Utility software that detects viruses by comparing files with a database of virus signatures, then removes or quarantines them; it must be updated regularly.
Credited: U1 2022 Q5

File fragmentation
When a file's data blocks are stored in separate, non-adjacent places on a disk because there is not enough continuous free space.

Task scheduler
A utility that shares processor time between tasks (e.g. round robin) and runs jobs such as backups, virus scans or updates automatically at set times.
Credited: U1 2019 Q5, 2022 Q5

Backup
A copy of data stored separately, preferably off-site, so it can be restored if the original is lost or damaged.
Credited: U1 2022 Q1, 2023 Q12, 2024 Q10

Full backup
A backup in which all the data is copied; it takes a long time and needs a lot of storage.

Virus
A harmful program that attaches itself to a file and spreads to other files and computers when the file is opened.

Quarantine
Isolating an infected file so it cannot run or spread, while keeping it in case it needs to be restored.

Exam focus

Utilities are tested often. In 2019 Q5(b) pupils had to describe what disk defragmenting and task scheduling do. For defragmenting, the marks come from three ideas: files become split into fragments across the disk, the defragmenter moves the fragments so each file is stored together, and this makes files faster to access. Writing only 'it tidies up the disk' will not score. For task scheduling, say that it shares processor time between tasks or that it runs jobs such as backups or virus scans automatically at set times. Utilities and security came up in 2022 Q5(b) and 2023 Q5(d). Backups appeared in 2022 Q1 and 2023 Q12(b), and freeing up disk space in 2023 Q12(c). Key-term matching in 2024 Q10 included backup. When explaining why antivirus must be updated, link the two ideas: new viruses keep appearing, and the software can only detect viruses whose signatures are in its database.

Unit 1 2024 Q10	[8]	Key term matching
Unit 1 2023 Q12	[4]	Batch processing, backups, freeing disk space
Unit 1 2023 Q5	[10]	CPU, system software resources, real-time vs multi-user, utilities, I/O devices
Unit 1 2022 Q5	[6]	3D printing, utilities and security, graphics digitiser
Unit 1 2022 Q1	[9]	MCQ: storage, bit rate, real-time, compression, buffering, ADC, big data, backup

? Check yourself

1. The operating system on John's computer includes a utility application. What is the function of the following two tasks carried out by utility software? (i) Disk defragmenting (ii) Task scheduling [4]
2. Explain why files become fragmented on a hard disk drive. [3]
3. Describe how disk defragmenting improves the performance of a computer. [3]

Answers: 1. (i) Defragmenting: files become split into fragments stored in different places on the disk (1) / the defragmenter rearranges the fragments so each file is stored in continuous blocks / free space grouped together (1). (ii) Task scheduling: divides processor time between tasks / users, e.g. round robin (1) / or runs jobs such as backups / virus scans automatically at set times (1). 2. The OS breaks files into blocks and stores them in any free space (1) / as files are saved and deleted, free space is left in small gaps spread across the disk (1) / so a new large file has to be split between several non-adjacent gaps (1). 3. Moves the blocks of each file so they are stored together / contiguously (1) / groups free space together (1) / the read/write head moves less, so files are read / saved more quickly (1).